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# Studies on application of scissor-jack braced viscous damper system in wind turbines under seismic and wind loads

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ABSTRACT

With the rapid development of the wind energy industry many wind farms have been constructed in regions prone to earthquakes and strong winds, which sometimes cause serious structural vibration problems for wind turbines. This paper details the development of a scissor-jack braced viscous damper system (VD-SJB) to suppress excessive vibration of the wind turbine tower. This system can be installed inside the tubular steel tower and is a suitable technology to enhance the structural performance of existing wind turbines. A detailed finite element wind turbine model was built and two damper systems with and without SJB were modelled and compared. Ground motions and strong lateral winds were applied as external loads to operational and parked turbines, respectively. Parameter optimization was performed by building a surrogate model using an artificial neural network and a multi-objective genetic algorithm. The resulting Pareto sets were discussed and the optimal system was evaluated to demonstrate the effectiveness of VD-SJB. Results show that both viscous dampers installed vertically and VD-SJB are able to reduce structural vibrations under seismic or wind conditions. However, VD-SJB is capable of magnifying the stroke of the damper and decrease the damping force, thus it is a more practical solution.

## 1. Introduction

The rate of installation of wind energy has increased greatly in recent decades, leading to a current global installed wind capacity of over 597 GW [1]. Due to this increasing demand for more plentiful wind resources, many wind farms have been built in wind or earthquake hazard-prone regions, such as the southeast coast of China. Additionally, recent developments in wind energy technology have led to a new generation of high capacity wind turbines (WTs) of much greater size. The support structure of a WT, traditionally a tubular steel tower, has low stiffness and damping which can lead to excessive vibrations. This in turn can affect the power generation and may lead to fatigue damage or serious collapse events, as have occurred several times in the past twenty years [2]. Therefore, it is necessary to develop effective damping devices that can suppress tower vibration to ensure the efficient power generation and structural safety of wind turbines.

Recent vibration control studies of WTs are summarized in Table 1. Most of these studies have focused on tuned-type dampers such as tuned mass dampers (TMD) [3–6], multi tuned mass dampers (MTMD) [7,8], hybrid mass dampers (HMD) [9], tuned liquid dampers (TLD) [10,11] or tuned liquid column dampers (TLCD) [12–15]. Comparatively less research has focused on non-tunable dampers such as magnetorheological dampers (MR) [16].

Tuned-type dampers generally require additional mass that aggravates the P-delta effect in the tower. They also lack robustness when external loading is applied over a wide frequency band, and WTs are generally subjected to multiple hazards including wind, earthquake and wave loading which contain prominently different frequency contents [19]. Additionally, tuned-type dampers may face installation problems when trying to retrofit in-service WTs. Thus, a promising alternative to tuned-type dampers are energy-absorbing viscous dampers as proposed in [17], which occupy little space, and have high applicability for wide frequency band processes.

The main objective of this paper is to study the use of energy-

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**Table 1**
Literature review of damper applications in wind turbine.

| Year | Authors | Device | Control technique | External loadings |
|---|---|---|---|---|
| 2015 | Caterino [16] | MR | semi-active | wind |
| 2015 | Chen et al. [13] | TLCD | passive | wind-wave and earthquake |
| 2015 | He et al. [3] | TMD | passive | wind-wave |
| 2016 | Ghaemmaghami et al. [11] | TLD | passive | seismic |
| 2016 | Brodersen et al. [17] | VD | active | wind-wave |
| 2016 | Zhang et al. [10] | TLD | passive | wind-wave |
| 2017 | Brodersen, et al. [4] | TMD | active | wind-wave |
| 2017 | Hu, et al. [9] | HMD | active | wind-wave |
| 2017 | Hussan, et al. [8] | MTMD | passive | seismic |
| 2017 | Zuo, et al. [18] | MTMD | passive | wind, wave and earthquake |
| 2017 | Sun [5] | TMD | semi-active | wind, wave and earthquake |
| 2017 | Sarkar, et al. [14] | TLCD | semi-active | wind |
| 2018 | Buckley, et al. [15] | TLCD | passive | impulse excitation |
| 2018 | Sun, et al. [6] | TMD | passive | wind-wave and earthquake |

absorbing dampers for wind turbine tower vibration control. The largest challenge of employing energy-absorbing dampers in WT towers is that the small tower deformations limit the damper stroke which bottlenecks the maximum damping effect. A possible solution to this problem is to use geometric amplification mechanisms such as toggle brace systems [20]. Toggle brace systems have been studied for two decades and modern incarnations include the open-space damper system by Polat and Constantinou [21–23]. Toggle brace systems have been studied comprehensively in frame structures [24,25] but only Brodersen et al. [26] have studied these systems in WTs.

This paper proposes a new toggle brace system: a scissor-jack braced viscous damper (VD-SJB) system for WTs, and details an analysis that compares it to VD without the brace as well as an undamped WT. To make the study more comprehensive and better reflect reality, multiple loading conditions were considered; the onshore WT in this study was subjected to an earthquake-wind coupled condition and a strong wind condition. Section 2 of this paper presents the FE model of a typical WT and external loadings for analysis. Section 3 proposes two types of damper systems for comparative analysis. Empirical design was initially used to determine ideal damper location and geometry, followed by the use of an ANN-produced model and a multi-objective genetic algorithm (NSGA-II) to select the optimal damping coefficients through a Pareto solution set. Finally, a detailed evaluation of the optimal damper systems was performed. Conclusions are drawn in Section 4.

## 2. Wind turbine modeling and loading scenarios

### 2.1. FE modeling of the wind turbine

The studied structure is a typical three-blade horizontal axis wind turbine tower [27,28] equipped with a 1.5 MW WindPACT turbine system [29]. A summary of the WT is presented here, while the full properties and details of the structure can be found in [27,28]. The tower is a cylindrical hollow single steel tube with the total height of 61.8 m and wheel hub height of 64.65 m. The bottom and top diameters are 4355 mm and 2955 mm respectively, and the wall thickness varies from 25 mm to 10 mm. The total weight of the tower is approximately 90 t. The tower tube is divided into several prefabricated welded sections, which are connected by bolted flanges at 13.39 m and 34.19 m from the base. Each blade has a total length of 32.8 m and a mass of 3.35 t. The eccentric distance of the centers of mass of the hub and the nacelle to the center of the tower section are 2.5 m and 1 m and their weights are 16.83 t and 60 t, respectively.

A fine finite element (FE) model was established for this 65 m tall wind turbine tower in ABAQUS (version 6.14). The modelling of the WT is shown in Fig. 1. A combination of shell and beam elements were used to build the WT in order to decrease degrees of freedom (DOFs) significantly and improve the calculation efficiency. The section of the tower where the damper systems were connected was modelled using a very fine mesh of 3D shell elements to capture the complex response. The remainder of the tower was modelled with coarse-mesh 3D beam elements connected to the shell elements using rigid coupling. Since previous WT studies generally use shell element FE models [27,28,30–32], a separate fine-meshed shell element model was also built and compared with the coupled shell-beam model. It was observed that the mean absolute percentage error (MAPE) between the two methods under external loadings presented in Section 2.2 was below 10%, so the accuracy of the coupled shell-beam model is acceptable. The damper systems were installed near the bottom of the tower, which is further discussed in Section 3.2. The turbine blades were modeled using 3D beam elements and generalized cross sections were used to match the actual blade section area, moment of inertia and polar moment of inertia. The mass of each blade element was lumped at a reference point. The hub and nacelle were represented as points of lumped mass, which were coupled to the top section of the tower by rigid connections. Additionally, a dashpot was set in the fore-aft direction between the lumped mass point at the hub and a fixed point to approximate the aerodynamic damping effect present in operational turbines. Its damping coefficient is derived in Section 2.2.3. The steel of the tower wall is S355 [27] and was modeled as a linear elastic-linear plastic material with a yield stress of 355 MPa, an ultimate stress of 470 MPa, an ultimate strain of 0.547, an elastic modulus of 210 GPa, a Poisson's ratio of 0.3, and a density of 7850 kg/m$^3$. The blades are made of fiberglass and only their elasticity is considered in the analysis with an elastic modulus of 21 GPa.

Modal analyses of the coupled shell-beam WT model and the shell model were performed to obtain the natural frequencies and mode shapes which were compared with field measurements of a real WT tower [33,34]. The results are compared in Table 2, where it can be seen that the modal frequency of the main shell-beam coupling model is close to the field measurement. Fig. 2 shows the first three flexural modes of the WT tower with blades. This shows that the coupling of blades and tower is obvious, especially for higher modes. Thus, this paper adopts such a model, even if many studies simplify the blades as a lumped mass point [27,28,30,31].

To accurately capture the performance of the turbine across all major modes as well as the damper motions designed later, the HHT implicit direct integration algorithm [35] was employed to perform the time history analyses.

### 2.2. Considered loads

Two external load conditions were considered: earthquake stimulation with service wind loads while the rotor is operational and strong wind condition when the rotor is parked, abbreviated as OPE and PAW respectively. The ground motion selection for OPE condition, wind

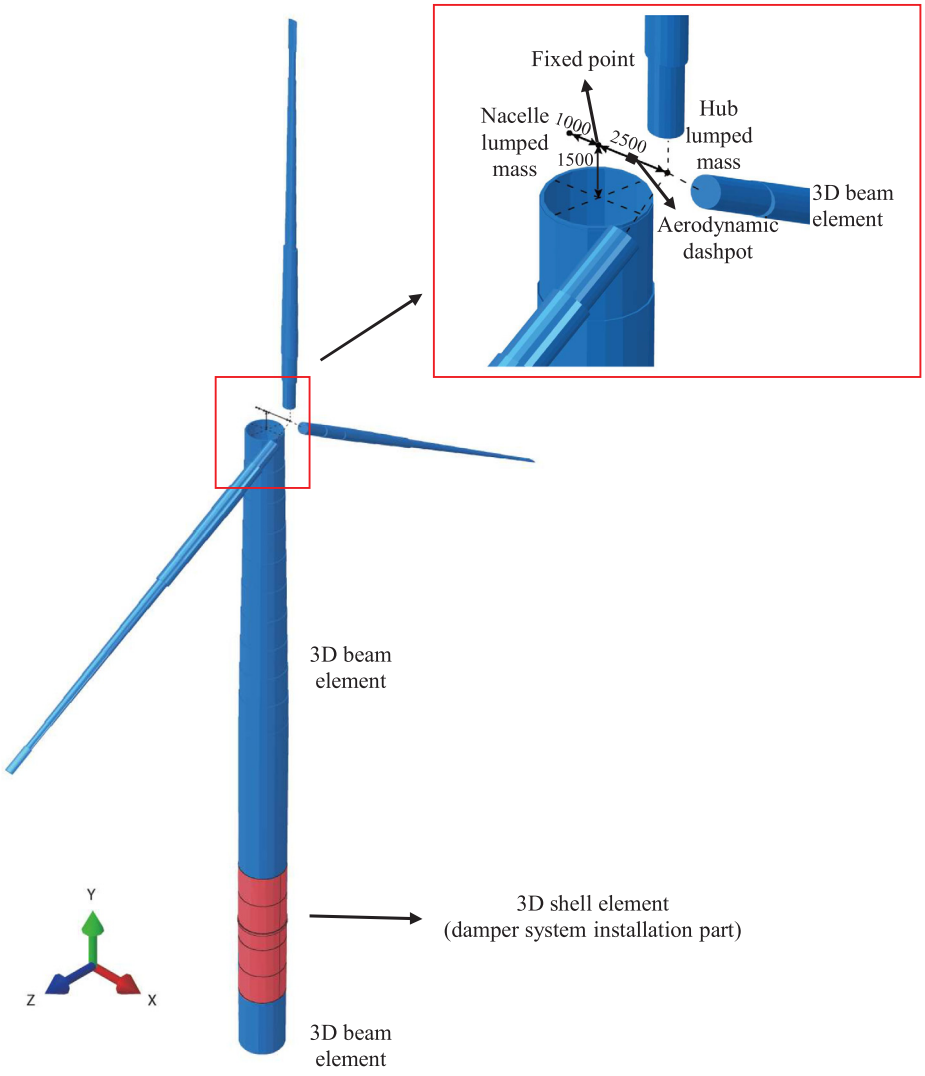



**Fig. 1.** FE model of WT.

profile generation for OPE and PAW conditions, aerodynamic damping simplification and code-to-code validation for OPE condition are discussed in the following sections.

### 2.2.1. Ground motion records

The ground motion records were selected from the PEER database of strong ground motions [36] to match the 5%-damping design response spectrum given in the Chinese seismic standard [37] after scaling. The maximum spectral acceleration of this design response spectrum (RS) is 0.9 g and the characteristic period $T_g$ is 0.4 s, which were determined based on the stiff site in which the field measurements were obtained for the WT. For the sake of more accurate RS matching, the shear velocity of the soil was included in the selection conditions. The information of selected ground motion records and their RS are shown in Fig. 3 and Table 3.

A total of seven ground motions recorded from different strong earthquakes were selected as required by the Chinese seismic standard [37]. All the records were scaled to 0.4 g peak ground acceleration (PGA). Though trial calculations, this wind turbine tubular steel structure would remain elastic under the 0.4 g PGA ground motion as the stress in the tower never exceeded the steel's yielding stress. However, geometric nonlinearity was considered in the ABAQUS model. The horizontal and vertical components of the ground motion record were applied since both affect the vertical deformation of tower wall. The baseline correction process from [38] was performed on the selected records. Since all the duration of seismic records is less than 60 s, the total duration of the seismic analysis under operational condition was 90 s where the ground motion was applied at 30 s.

### 2.2.2. Aerodynamic loading generation

FAST [39], a multi-body dynamics-based WT analysis software by NWTC Information Portal, was used to generate the aerodynamic

**Table 2**
Natural frequencies of the tower and error in X direction.

| WT | 1st mode frequency/Hz | Error/% | 2nd mode frequency/Hz | Error/% |
|---|---|---|---|---|
| Actual field measurement | 0.49 | – | 3.84 | – |
| Shell model | 0.49 | 0 | 4.32 | 13 |
| Coupled shell-beam model | 0.49 | 0 | 4.19 | 9 |

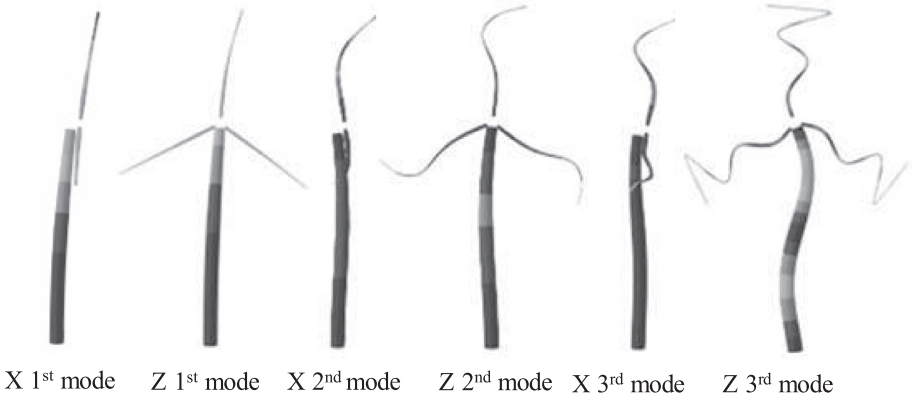



**Fig. 2.** The first three flexural modes of the wind turbine tower.

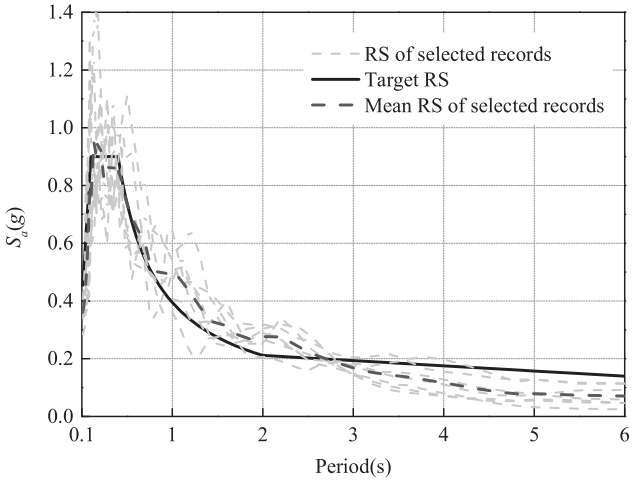



**Fig. 3.** Ground motion acceleration response spectra.

loading at the top of the turbine tower including the blade loads. A combined FAST-ABAQUS approach to obtain aerodynamic loading (where ABAQUS operates as the post-process of FAST) has been widely adopted in previous studies [32,40]. The procedure to obtain the top loading is as follows:

(a) generate artificial wind profiles using TurbSim [41];
(b) build a FAST model based on WT information and perform dynamic analysis under generated wind profiles;
(c) extract time history of top loading (force and moment) and apply to the hub position of the ABAQUS model.

Wind loads along the tower wall and at the nacelle were generated using Eq. (1) from the Chinese load standard [42]:

$$F(h) = 0.5\rho v(h)^2 A_p(h) C_d \tag{1}$$

where $A_p(h)$ is the projected area of tower section of height $h$; $C_d$ is the drag coefficient: 1.2 for the tower and 1.3 for the nacelle [42]; ρ is the air density; and $v(h)$ is the velocity of the reference point at height $h$. The wind profile is assumed as an exponential distribution with an exponent of 0.2 [43].

Two WT rotor conditions based on wind speed were considered as previous mention. When the hub wind velocity is 11.5 m/s – the rated wind velocity of the turbine [29] – the WT is under the operational condition. When the hub wind velocity is 42.5 m/s – the extreme design velocity of the WT [43] – the WT is in the parked condition. Wind loading in the operational condition is parallel to the rotor axis, whereas loading for the parked condition is perpendicular to the rotor axis which is the worst case scenario when the blades are feathered [23]. Following WT design standards [43], the normal and extreme turbulence models were used for the operational and parked conditions, respectively. Generally, the duration of wind-resist analyses is over 600 s, however 90 s analyses were used in this study for both wind and seismic loading due to the large number of calculations required during the optimization process. Three different wind profiles were generated for the PAW condition considering the stochasticity of wind. Since earthquake loading mainly governs in the OPE condition, just one service-level wind profile was simulated and applied.

The power spectrum density (PSD) of the top loading obtained by FAST is shown in Fig. 4. It can be seen that for the operational condition, there are peaks at approximately 1 Hz, 2 Hz and 3 Hz which corresponds to the 1 time, 2 times and 3 times the blade passing frequency of this WT (the rotor rotation speed is around 20 rpm [29]). In the parked condition, these three frequency components disappear since the rotor speed is zero. The PSDs of both the operational and parked condition contain a 3–4 Hz frequency peak, which results from the natural frequency of the blades.

### 2.2.3. Aerodynamic damping

The seismic analysis under operational condition should also consider the interaction of the rotating rotor and external winds, which is usually simplistically simulated by an aerodynamic damping ratio. Current studies [19,32,40,44] generally apply a total damping ratio of 5% to models of operational WTs. However, using material damping in

**Table 3**
Selected ground motion records.

| No. | Earthquake name | Year | Station | Magnitude | $V_{s30}$ (m·s$^{-1}$) |
|---|---|---|---|---|---|
| 1 | Imperial Valley-02 | 1940 | El Centro Array #9 | 6.95 | 213.44 |
| 2 | Superstition Hills-02 | 1987 | Westmorland Fire Sta | 6.54 | 193.67 |
| 3 | Chi-Chi Taiwan | 1999 | TCU122 | 7.62 | 475.46 |
| 4 | Manjil Iran | 1990 | Abbar | 7.37 | 723.95 |
| 5 | Iwate Japan | 2008 | IWT010 | 6.9 | 825.83 |
| 6 | Imperial Valley-06 | 1979 | El Centro Array #12 | 6.53 | 196.88 |
| 7 | Darfield New Zealand | 2010 | Christchurch Cashmere High School | 7 | 204 |

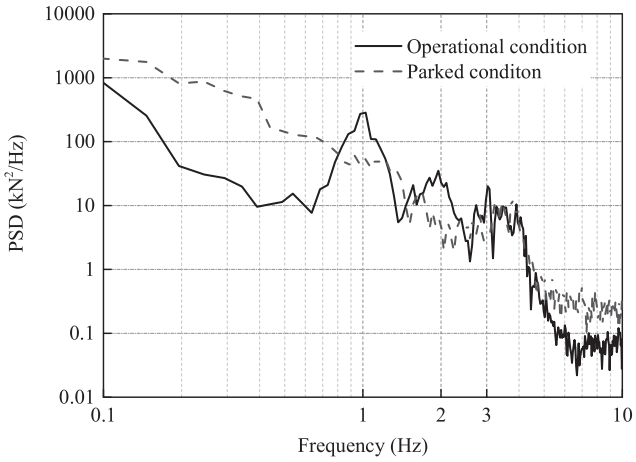



**Fig. 4.** Aerodynamic loading PSD.

this way overestimates the damping effect in the side-to-side direction (which in reality is close to 1%) and can lead to an underestimated response. To solve this problem, this study proposes using an aerodynamic dashpot in the FE model to add one-directional damping parallel to the rotor axis, as shown in Fig. 1. Since the spinning blades are the source of the aerodynamic damping, the dashpot was placed at the hub height. The damping coefficient value of the aerodynamic dashpot was solved using the closed-form equation proposed in [45]. Since the dimensions of the blades in [45] are the same as this study but the stiffness and mass of the tower are not, the equivalent damping ratio is calculated to be 2.8% rather than the proposed 4%.

In order to verify the validity of this aerodynamic dashpot simplification method, a blade-tower-nacelle integrated model of the study WT was built in FAST and analyzed using a seismic module [46] capable of performing coupled analysis of aerodynamic and seismic loads [47]. The Seismic Record 1 and the rated hub wind velocity under operational condition were applied during validation analysis. This FAST model was compared to ABAQUS models employing the two aerodynamic simplification methods: the material damping method using 3.8% structural damping and the aerodynamic dashpot with 1% material damping plus the 2.8% dashpot.

Fig. 5 shows the response time histories of the relative displacement of the top of the tower in both horizontal directions. It can be seen that in the fore-aft direction the response of the three models is similar, which means the dashpot simplification method can effectively simulate aerodynamic damping effect. The root-mean-square (RMS) error for the material damping method and aerodynamic dashpot method compared to the FAST model was 4% and 3%, respectively. However, in the side-to-side direction, the material damping method inaccurately reduced the response as it added nonexistent aerodynamic damping in this direction. The dashpot method proves much more accurate by comparison; the RMS error for the additional material damping method and aerodynamic dashpot method was 58% and 11%, respectively.

## 3. Viscous damper system design

### 3.1. Magnification effect of scissor-jack braces

As mentioned in Section 1, the largest challenge of employing viscous dampers in WT towers is that the small tower deformations limit the damper stroke, so the damping force should be very large to reach the ideal damping effect, which is not practical in application. Thus, a scissor-jack braced system for stroke-amplification for use in WTs is proposed. Sigaher et al. [48] originally proposed scissor-jack braces as a system for magnifying the effect of dampers under seismic loading. Since then, several researchers [24,49] have analyzed this kind of brace and proved its effectiveness in structural vibration mitigation and performance enhancement. The basic form of this SJB is shown in Fig. 6.

With the assumption of an infinite stiffness brace, it can be obtained from equilibrium of force and displacement coordination relation that:

$$F = F'cot\frac{\theta}{2} \tag{2}$$

$$D' = Dcot\frac{\theta}{2} \tag{3}$$

where $\cot\frac{\theta}{2}$ corresponds to the magnification factor, henceforth named $f = \cot\frac{\theta}{2}$.

Considering the damper is a velocity-relative viscous damper and obedient to a nonlinear exponent model with damping exponent $\alpha$:

$$F' = C\,|\dot{D}'|^{\alpha} sign(\dot{D}') \tag{4}$$

Since velocity $\dot{D}'$ is the derivative of displacement $D'$ about time, the magnification factor of velocity is also $f$. Assuming that the intersection angle $\theta$ is invariant over time, Eq. (5) can be obtained:

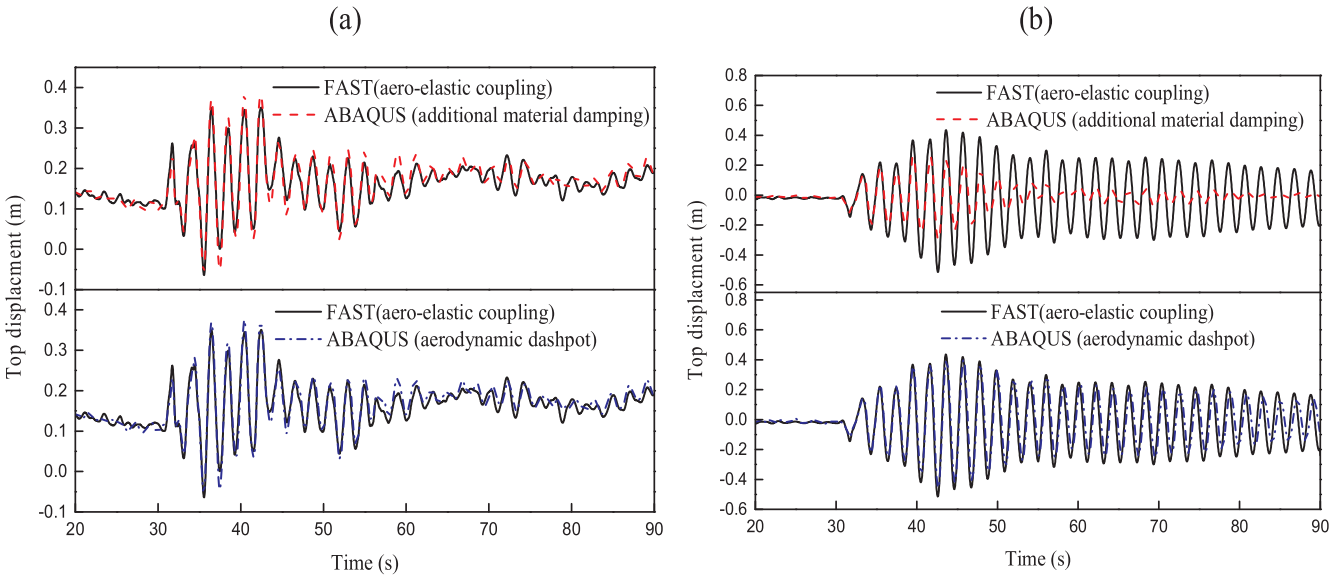



**Fig. 5.** Top displacement comparison in (a) fore-aft direction; (b) side-to-side direction.

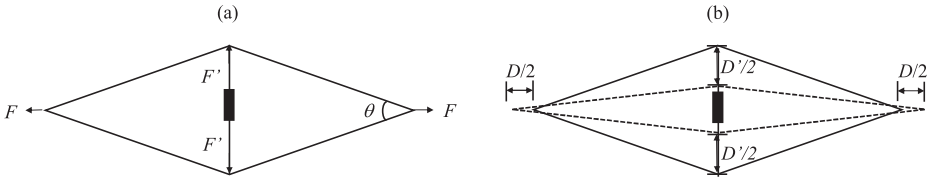



**Fig. 6.** Force diagram and displacement coordination of SJB device.

$$F^{'} = C\ |\dot{D}'|^{\alpha} sign(\dot{D}^{'}) = Cf^{\alpha}\ |\dot{D}|^{\alpha} sign(D) \tag{5}$$

By substituting Eq. (5) to Eq. (2), the constitute relationship of VD-SJB is found to be:

$$F = fF^{'} = Cf^{1+\alpha}\ |\dot{D}|^{\alpha} sign(\dot{D}') \tag{6}$$

Comparing Eq. (4) with Eq. (6), it can be seen that the magnification effect of the damping force is $f^{1+\alpha}$. Factor $f$ is a function of $\theta$ that will be affected by the vertical deformation of the tower wall, the deformation of the brace, and the force from the damper. The damping exponent $\alpha$ also affects the magnification effect nonlinearly. Thus, the VD-SJB was finely modelled in FE model to simulate the nonlinearity accurately. An additional viscous damper system without SJB was also modelled and compared to the VD-SJB to study the stroke amplification effect of this system.

### 3.2. Empirical design of damper layout and FE modelling

#### 3.2.1. Layout height design

In order to place the damper systems into a reasonable location the displacement shape of the WT was extracted, which is shown in Fig. 7. It can be seen that the absolute displacement angle increases with height, as expected of a general bending-dominant structure. However, according to previous studies [50], the stroke of the vertical dampers is dependent on the "harmful displacement angle" which can be calculated by several methods discussed in [51]. In this study, the "improved secant method" formulated in Eq. (7) was adopted to calculate harmful displacement angle $\Delta\beta_i$ at the $i_{st}$ tower element

$$\Delta\beta_i = \beta_{i+1} - \beta_i \tag{7}$$

where $\beta_i$ and $\beta_{i+1}$ are the displacement angles at the $i_{st}$ and $i_{st+1}$ of tower elements, respectively. As the Fig. 7 shows, there are two sudden changes of the harmful displacement angle which are induced by high-stiffness flanges. In general the harmful displacement angle decreases with height, however since the lowest section of the tower has increased wall thickness to account for the door opening, the harmful displacement angle is largest at a height of 6 m from the base of the tower. This behavior is discussed in some studies [50,51] and is also supported by the observation that collapse events from wind or earthquake loading usually occur at or near the bottom of the WT [2,28,30]. Reducing rotation at the base of the WT will also result in greater suppression of displacements at the top of the tower due to the larger moment arm. Therefore, the damper systems were placed at the bottom part of the WT (from 6 m to 20 m above the ground as shown in Fig. 1) in this study.

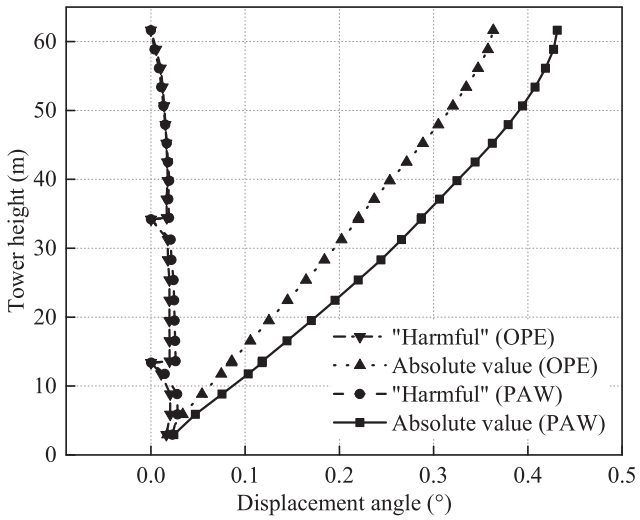



**Fig. 7.** Displacement angle along the WT height.

#### 3.2.2. Layout configuration and selection of number of dampers

In traditional frame structures, VD-SJBs and VDs are typically installed diagonally or obliquely [48,52]. However, in this study the damper systems were instead installed in the vertical direction to maximize the damping effect due to the unique behavior of wind turbines [17]. Since WT towers are tall flexible structures with relatively low natural frequencies, the bending deformation is dominant (rather than the shear deformation), which manifests as tension or compression strain along the tower wall. Trial calculations revealed that the available vertical deformation was several times greater than the available diagonal deformation.

The selection of the number of viscous dampers within the VD-SJB system is an important design consideration, which includes selecting the number of layers of the system and number of dampers within each layer. Firstly, designers can refer to harmful displacement angle distributions (shown in Fig. 7) to determine the height range of the tower to install dampers. Considering the base height of a single layer and the available budget, the appropriate number of layers to be built can be determined. The number of dampers within one layer should be selected and validated by verifying that local stress concentrations remain below material yielding stress. To account for the two-dimensional nature of the loading on a wind turbine, no less than three dampers should be employed and they should be oriented symmetrically around the turbine tower.

In this study, according to this empirical design criteria a two-layer damper system with approximate heights of 6 m and 8 m, respectively, was installed, with each damper layer consisting of four vertically oriented viscous dampers.

#### 3.2.3. FE modeling of damper systems

The layout of the VD-SJB and VD systems are shown in Fig. 8. These systems suspend the vertically oriented dampers between ring stiffeners, which are used to reduce stress concentration but do not notably impact the stiffness or natural frequency of the overall structure. In the numerical FE model, the stiffening rings of the VD-SJB were modelled using rigid body shell elements. The steel SJB members were modelled using beam elements with 150 mm diameter circular cross-sections and an intersection angle $\theta$ of 10°. The brace-brace and brace-stiffener connections were modelled as hinges that restricted out-of-plane DOFs. The viscous damper was modelled using a nonlinear dashpot element. For VD without braces, the dampers were directly modelled using nonlinear dashpot elements pinned at the top and bottom to stiffening rings.

### 3.3. Optimization analysis design of damper coefficients

After empirically determining of the damper system layout, the next

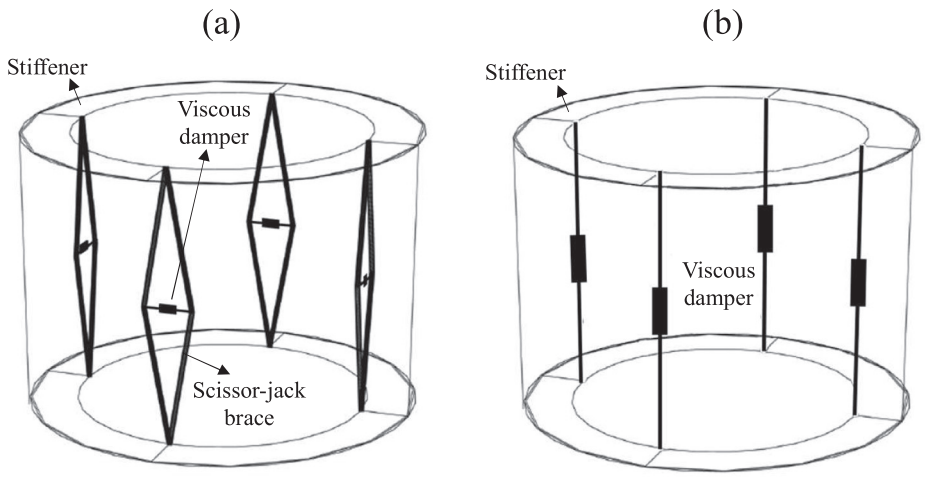



**Fig. 8.** Damper systems of (a) VD-SJB and (b) VD.

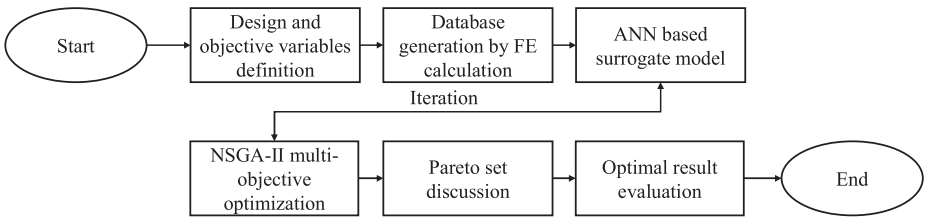



**Fig. 9.** Optimization flow chart.

step is to optimize the viscous damper coefficients. Numerical optimization was adopted in this study to accomplish this. The optimization flow chart is shown in Fig. 9. Each step is discussed in the following sections.

Firstly, the design and objective variables were defined. Since a detailed FE model was used to simulate the response of the WT, employing this model directly would be extremely time consuming for performing traditional direct parameter optimization. Instead, surrogate models [53,54] were constructed which were able to significantly reduce the overall computational cost by allowing the use of parallel computing as well as eliminating the need for the direct integration of FE calculations within the optimization process. Additionally, detailed response surfaces of the design variables were obtained, which are not produced by direct iteration optimization, and provide useful information for study. An artificial neural network (ANN) [54] was adopted to generate surrogate models based on a database of results from the coupled shell-beam FE model. The NSGA-II optimization algorithm [55] was utilized to obtain the Pareto set, which can be used to select the optimal solution based on the engineering demand. The NSGA-II algorithm is robust and converges quickly, thus it is frequently applied in engineering optimization [56,57]. Finally, the optimal result was evaluated using the FE model again for accuracy validation and assessment of damper performance. Optimization of the WT with VD-SJB and VD were performed in parallel then compared.

### 3.3.1. Design and objective variables definition

To begin, design variables and objective variables are selected. The damper coefficient ($C$ and $\alpha$) are the design variables used in this study. As previous discussion, the main issue of applying energy-absorbing damper in WTs is that the damper stroke is limited, thus, the optimal damper design is to generate the desired damping effect with the lowest damping force possible the maximize the damper stroke. In order to achieve this optimized design demand the double-objective functions ($Y_1$ and $Y_2$) are defined as:

$$Y_1(C, \alpha) = \frac{E(A_{\text{Damper}})}{E(A)} \tag{8}$$

$$Y_2(C, \alpha) = \mathrm{E}(B_{\text{Damper}}) \tag{9}$$

where $A_{\text{Damper}}$ and $A$ are the algebraic mean value of relative spatial displacement at the hub ignoring base movement with and without damper systems. Thus, $Y_1$ represents the mean response reduction ratio. The absolute mean value of the damping force of all the dampers is selected as $B_{\text{Damper}}$, whose variation tendency is opposite to the stroke of the damper. Due to the static wind effect the structural response has a non-zero mean, however in vibration control studies the fluctuating component of the response is the primary concern. Therefore, only the fluctuating components of the time history responses were considered during optimization. Due to variations in the indexes from record-to-record uncertainty, the arithmetic average operator E was adopted. Thus, the average result from the seven seismic records in the OPE condition and the average result from the three wind profiles in the PAW condition were used to calculate the multi-objective functions.

### 3.3.2. Database generation and surrogate models

A dynamic implicit solver was adopted to calculate the response time history. The HHT method with automatic integral step was used to solve the equations of motion. The optimization database was generated using a restricted range of variables; the range of damping exponents $\alpha$ was 0.4–1 for nonlinear viscous dampers [58]. The range of possible damping coefficients ($C$) is relatively broad, which can be customized by damper companies. To confirm a reasonable range of $C$ that avoids superfluous calculations, an initial trial-and-error analysis was performed to define the suitable range of $C$. Based on the trial calculation results, a range of $C$ of $2 \times 10^6$ to $20 \times 10^6$ N s/m was chosen for the VD-SJB, and for VD the range of $C$ was $100 \times 10^6$ to $1000 \times 10^6$ N s/m. The damping coefficient range of VD was greater than the range of VD-SJB due to the damping force magnification from the SJB shown in Eq. (5). The sampling interval of $\alpha$ was 0.1 and the sampling interval of $C$ was $2 \times 10^6$ N s/m and $100 \times 10^6$ N s/m respectively, which were chosen to balance generation accuracy and

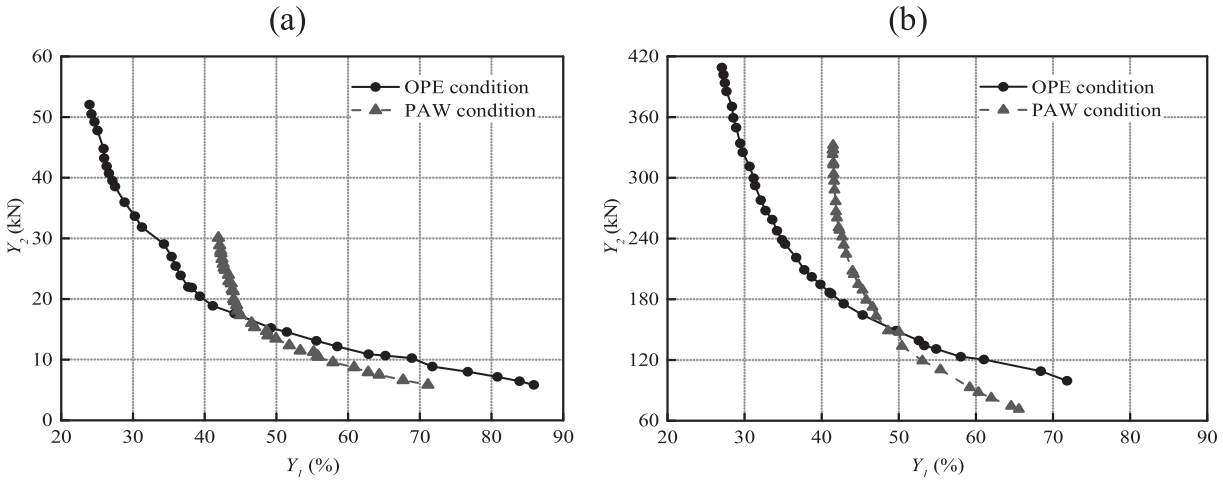



**Fig. 10.** Pareto sets of wind turbines equipped with: (a) VD-SJB (b) VD.

efficiency.

ANN fitting was adopted to train the ANN-based surrogate model. 70% of the sample runs were used to train the ANN while the remaining 30% were used to validate and test the ANN model. The validation runs were ensured to have a coefficient of determination $R^2$ above 0.99 compared to the ANN model, which shows that the ANN model harbored sufficient precision to surrogate the generated database.

### 3.3.3. Pareto set discussion

ANN-based surrogate models were used to calculate the objective functions Eq. (8) and Eq. (9) and the NSGA-II algorithm was adopted to perform optimization. The Pareto sets of the WT with VD-SJB and VD under OPE and PAW conditions are shown in Fig. 10. Recall that the mean reduction ratio ($Y_1$) and mean damping force ($Y_2$) were defined as the objective functions. The results show that they are conflicting objectives and that $Y_1$ has diminishing returns as $Y_2$ increases. Additionally, the dampers are more effective at reducing displacement under OPE compared to PAW conditions. While the vibration reduction effect of the VD and the VD-SJB are similar since their damping mechanism is the same, the VD-SJB is able to reach the same reduction with a lower damping force which means the same damping effect can be achieved by using a smaller damping coefficient and larger damper stroke. Practically speaking, when there is a specific displacement reduction demand, the lowest acceptable damping coefficient should be used to minimize the local stress and improve the producibility of the dampers.

Figs. 11–12 show the Pareto set points on objective variables contours. The distribution of Pareto set points is similar under OPE and PAW conditions, thus, only the results under OPE condition are presented here. Eq. (6) shows that the damping force magnification effect of the SJB is nonlinear, which is observed in Fig. 11(a) and Fig. 12(a). Based on the gradients of the figures, it can be concluded that for VD-SJB $\alpha$ is the more sensitive objective variable, thus it is better to increase $\alpha$ firstly and then increase $C$ for the ideal damping effect. By contrast, the VD-equipped WT is relatively independent of $\alpha$, therefore, increasing $C$ is recommended to obtain an improved damping effect.

### 3.3.4. Optimal damper evaluation

Assuming the mean reduction ratio demand $Y_1$ is 50%, the optimal damping coefficient $C$ and damping exponent $\alpha$ were selected based on the Pareto sets presented above, which is summarized in Table 4. In order to study the damping effect with greater detail and verify the accuracy of the ANN-based surrogate models, FE analysis was performed again using these optimal damper parameters. The mean reduction ratio ($Y_1$) was calculated based on Eq. (7). It can be seen that the direct actual $Y_1$ calculated by FE model is quite close to the $Y_1$ predicted by the surrogate model. Therefore, the precision of the ANN-based surrogate models is sufficiently fine to replace the FE model when performing optimization analysis. Additionally, it was found that the local stress maximums of all the cases was less than the yield limitation of the WT tower (355 MPa), thus local yielding from either damper wasn't a risk.

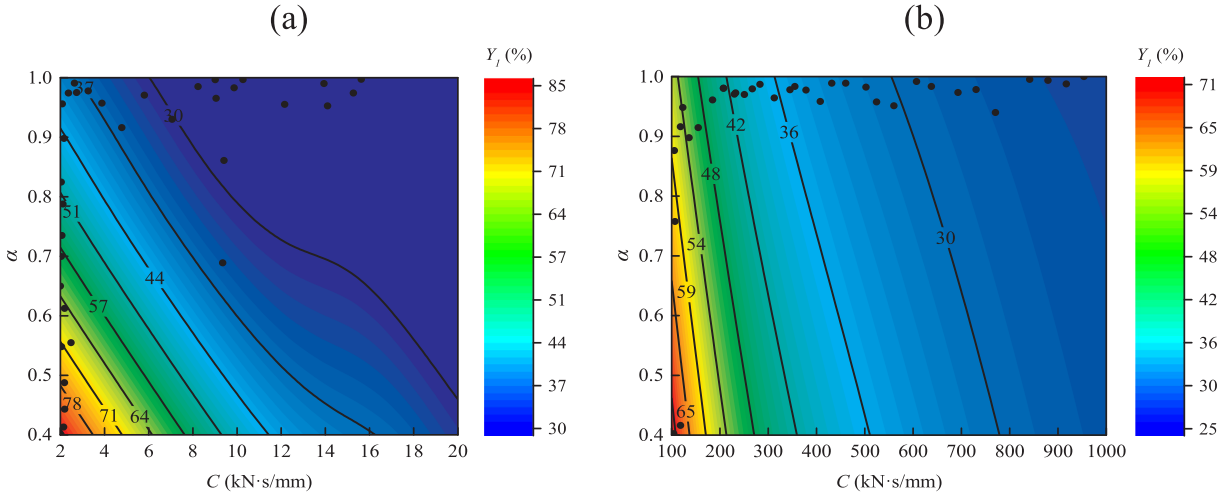



**Fig. 11.** Pareto set points on $Y_1$ contours under OPE condition: (a) VD-SJB; (b) VD.

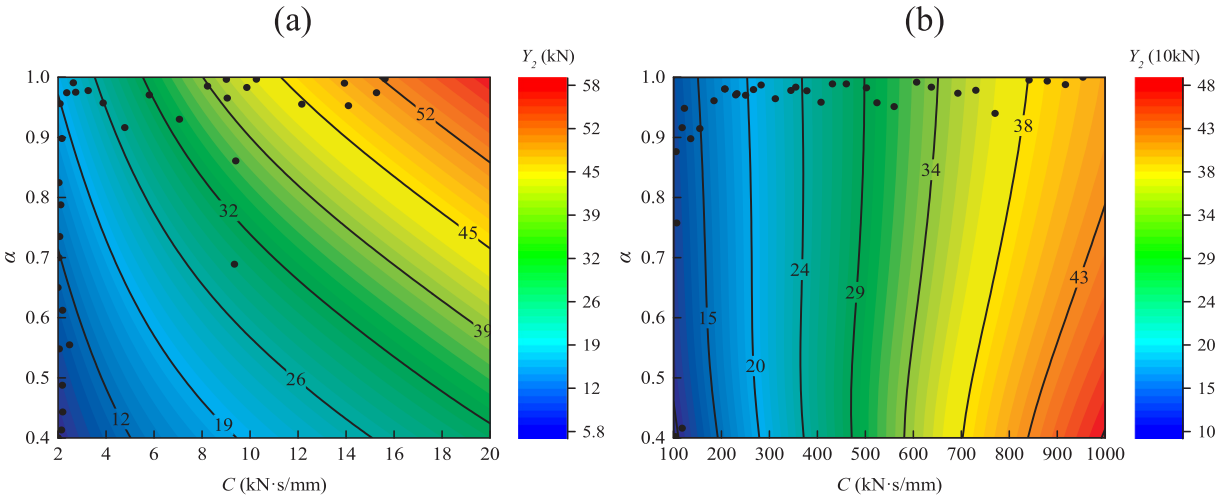



**Fig. 12.** Pareto set points on $Y_2$ contours under OPE condition: (a) VD-SJB; (b) VD.

**Table 4**
Optimal parameter of VD-SJB and VD.

| Damper | External condition | $C$ (N·s/m) | $A$ | Predicted $Y_1$ (%) | Actual $Y_1$ (%) |
|---|---|---|---|---|---|
| VD-SJB | OPE | $2.041 \times 10^6$ | 0.825 | 50 | 50.5 |
| VD | OPE | $155.09 \times 10^6$ | 0.914 | 50 | 49.2 |
| VD-SJB | PAW | $5.067 \times 10^6$ | 0.749 | 50 | 48.2 |
| VD | PAW | $351.30 \times 10^6$ | 0.829 | 50 | 49.0 |

Fig. 13 and Fig. 14 show two representative vibration reduction time histories (Seismic Record 1 and Wind Profile 1) of the optimal damper under OPE and PAW condition. It can be seen in these figures that both damper systems reduce the dynamic response of the WTs. Comparing Fig. 13(a) and (b), it can be seen that due to the presence of fore-aft aerodynamic damping provided by the operational blades, the vibration reduction effect of the damper system in the fore-aft direction is less than in the side-to-side direction. Additionally, it can be observed that the response time histories of WTs equipped with the VD-SJB and the VD are quite similar, which matches the previous observations about Fig. 10.

Like all displacement magnification mechanisms, the SJB can be sensitive to geometric changes. A representative time history of the SJB's intersection angle $\theta$ is shown in Fig. 15 to study the change of $\theta$ along time history, which was extracted from the bottom VD-SJB in the positive X direction of the FE model under OPE condition (Seismic Record 1). It can be seen that due to the large dimensions of the proposed VD–SJB that the range of variation of $\theta$ during operation is less than 1° and any geometric changes in the damper are small, thus the magnification factor $f$ (Eq. (2) and Eq. (3)) and the damping performance should not be influenced too much in this design case. In practical engineering, it must be noted that large errors in the SJB

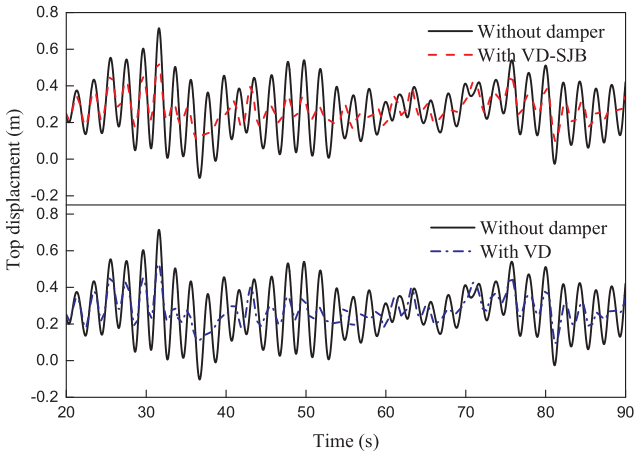



**Fig. 14.** Time history comparison under PAW condition (inflow wind direction of Wind Profile 1).

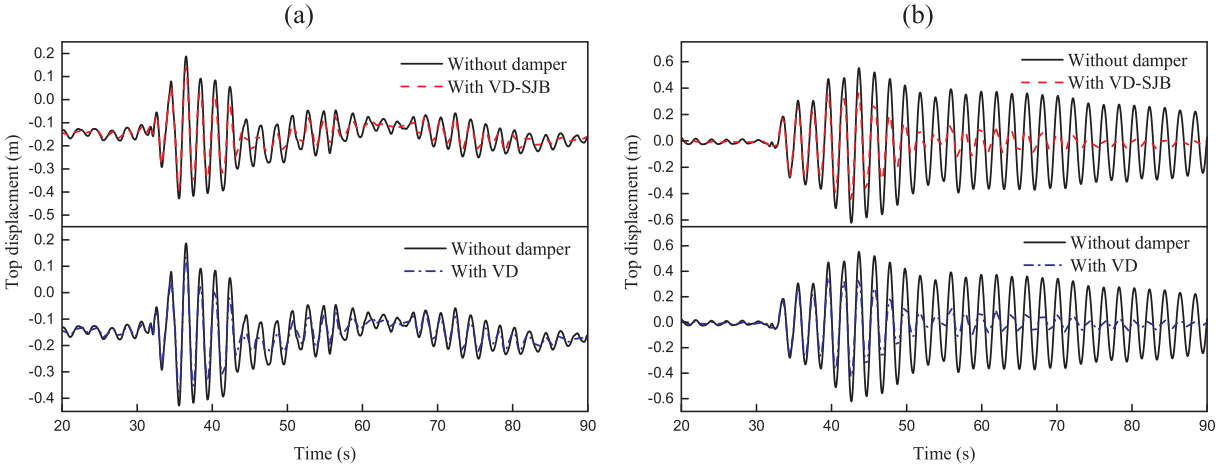



**Fig. 13.** Time history comparison under OPE condition (Seismic Record 1): (a) X direction; (b) Z direction.

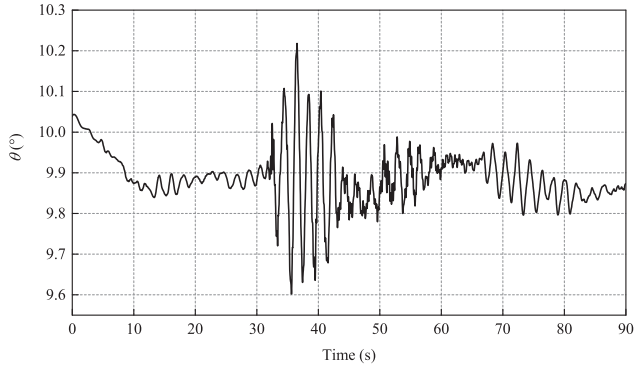



**Fig. 15.** $\theta$ variation of a representative VD-SJB under OPE condition (Seismic Record 1).

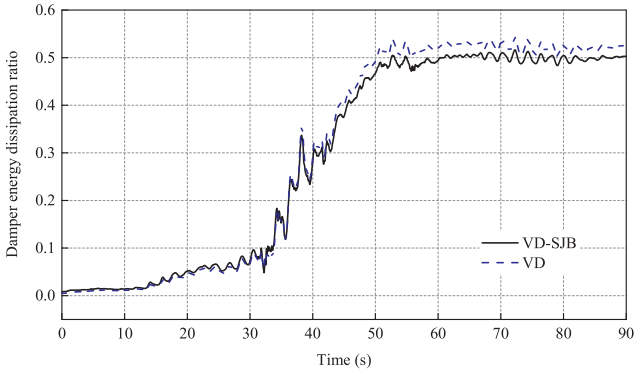



**Fig. 17.** Representative dissipated energy time history under VD-SJB and VD.

intersection angle will decrease the effectiveness of the damping system, so careful construction and installation of the VD-SJB system is required.

Furthermore, two representative hysteretic curves of the WT equipped with VD-SJB and VD are shown in Fig. 16. The hysteretic curve comes from the damper installed at the bottom of side-to-side direction under Seismic Record 1. It can be seen that the SJB can magnify the stroke (the maximum is magnified from approximately 2–24 mm) and correspondingly reduce the required damping force from the dampers. This stroke increase is particularly beneficial here as the small stroke of the VD without the brace may be below the required minimum stroke length for many industrial dampers. Correspondingly, damper energy dissipation ratio (defined as damper energy dissipation divided by external work) is shown in Fig. 17. It can be observed that with a larger stroke and less damping force the VD-SJB system has similar energy dissipation capacity as the VD system.

## 4. Conclusions

The application of VD-SJB in WTs was studied by optimizing the damper parameters using the ANN-GA method then studying the results using a fine FE simulation. The following conclusions can be drawn from this research:

1. In WT FE modeling, the dashpot element can be used to efficiently simulate the aerodynamic damping effect of operational turbines. Using this method, the different damping ratios of different directions can be accurately simulated.

2. A combination of surrogate models and optimization algorithms is able to separate time-consuming FE calculation from the optimization iteration step, which notably reduces the total optimization solving time. For structural engineering optimization problems, this strategy is suggested when models require very high detail or there is a large amount of data.

3. Utilizing the vertical deformation of the tower wall, VD or VD-SJB are able to effectively reduce the WT vibration equally under both earthquake and wind conditions. Under earthquake condition with operational rotor, due to the aerodynamic damping effect in the fore-aft direction, the mitigation effect of the dampers in the side-to-side direction is comparatively greater.

4. VD-SJB are not only able to magnify the stroke of damper to practical levels but also can decrease the required damping force to obtain the same damping effect compared to brace-less VD. Thus, the implementation of VD-SJB appears to be a practical engineering solution for the problem of excessive wind turbine vibrations.

Additionally, the proposed VD-SJB system can be further explored. The performance-based response mitigation study is a promising direction. The code-based practical optimization design is an imperative development direction as well. Experimental study is also necessary to test the applicability and practicality of the VD-SJB system applied in WTs.

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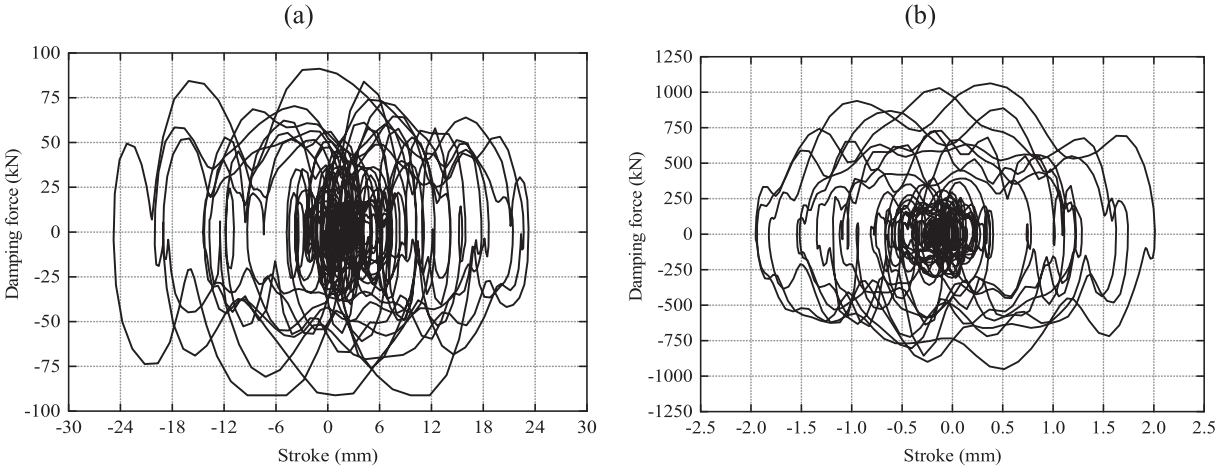



**Fig. 16.** Representative damper hysteretic curve under (a) VD-SJB; (b) VD.

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